Practice workbook · Area of a triangle or quadrilateral from coordinates

Finding the area of a triangle, trapezium, parallelogram, kite or other quadrilateral whose vertices are known, using base and perpendicular height, half the product of perpendicular sides or diagonals, splitting into triangles, or the coordinate (shoelace) method.

17 question parts: 1 worked-example retry and 16 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-COO-09

1. Question 6 · part c · 4 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part c only.

6 The point A has coordinates (3, 0) and the point B has coordinates (2, 2).
The line L_1 passes through B and is perpendicular to AB.

(a) Find an equation of L_1

Give your answer in the form $ax + by + c = 0$

(5)

The line L_2 with equation $x - 7y - 3 = 0$ intersects the line L_1 at the point C.
The midpoint of AC is M.

(b) Find the coordinates of M.

(5)

(c) Find the area of the triangle ABM.

(4)

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1. Question 6 · part c · 4 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part c only.

Question 6 continued

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1. Question 6 · part c · 4 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part c only.

Question 6 continued

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1. Question 6 · part c · 4 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part c only.

Question 6 continued

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(Total for Question 6 is 14 marks)

1. Question 6 · part c · 4 marks · Working and answer

Working and answer

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2. Question 11 · part e · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part e only.

11 The points A and B have coordinates $(-1, 3)$ and $(5, 6)$ respectively.

(a) Find an equation for the line AB.

(2)

The point P divides AB in the ratio $2:1$

(b) Show that the coordinates of P are $(3, 5)$

(2)

The point C with coordinates (m, n), where $m > 0$, is such that CP is perpendicular to the line AB.

Given that the radius of the circle which passes through A, P and C is 5

(c) find the value of m and the value of n.

(6)

The point D with coordinates (p, q) is such that the line AD is perpendicular to the line AB and the line DC is parallel to the line AB.

(d) Find the value of p and the value of q.

(3)

(e) Find the area of trapezium $ABCD$.

(4)

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2. Question 11 · part e · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part e only.

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Question 11 continued

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2. Question 11 · part e · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part e only.

Question 11 continued

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2. Question 11 · part e · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part e only.

Question 11 continued

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2. Question 11 · part e · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part e only.

Question 11 continued

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(Total for Question 11 is 17 marks)

2. Question 11 · part e · 4 marks · Working and answer

Working and answer

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3. Question 5 · part c · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

5 The points P, Q, R and S have coordinates (4, 7), (3, 0), (10, 1) and (11, 8) respectively.

(a) Show, by calculation, that the lines PR and QS are perpendicular.

(3)

(b) Find the exact lengths of (i) PR (ii) QS

(2)

(c) Find the area of the quadrilateral $PQRS$

(2)

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3. Question 5 · part c · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

Question 5 continued

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(Total for Question 5 is 7 marks)

3. Question 5 · part c · 2 marks · Working and answer

Working and answer

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4. Question 10 · part d · 3 marks

2022 JAN · Paper 01

Further practice · Attempt part d only.

10 The equation of the line L_1 is $y - 2x - 6 = 0$

The point P with coordinates $(2, a)$ lies on L_1

(a) Find the value of a

(1)

The line L_2 is perpendicular to L_1 and passes through P

(b) Show that an equation of L_2 is $x + 2y - 22 = 0$

(4)

Line L_1 crosses the x-axis at the point A and line L_2 crosses the x-axis at the point B

The point C has coordinates (m, n) such that $m > 0$ and $n < 0$

The length of AC is $5\sqrt{2}$ and the gradient of BC is $\frac{1}{4}$

(c) Find the value of m and the value of n

(9)

(d) Find the area of quadrilateral $ACBP$

(3)

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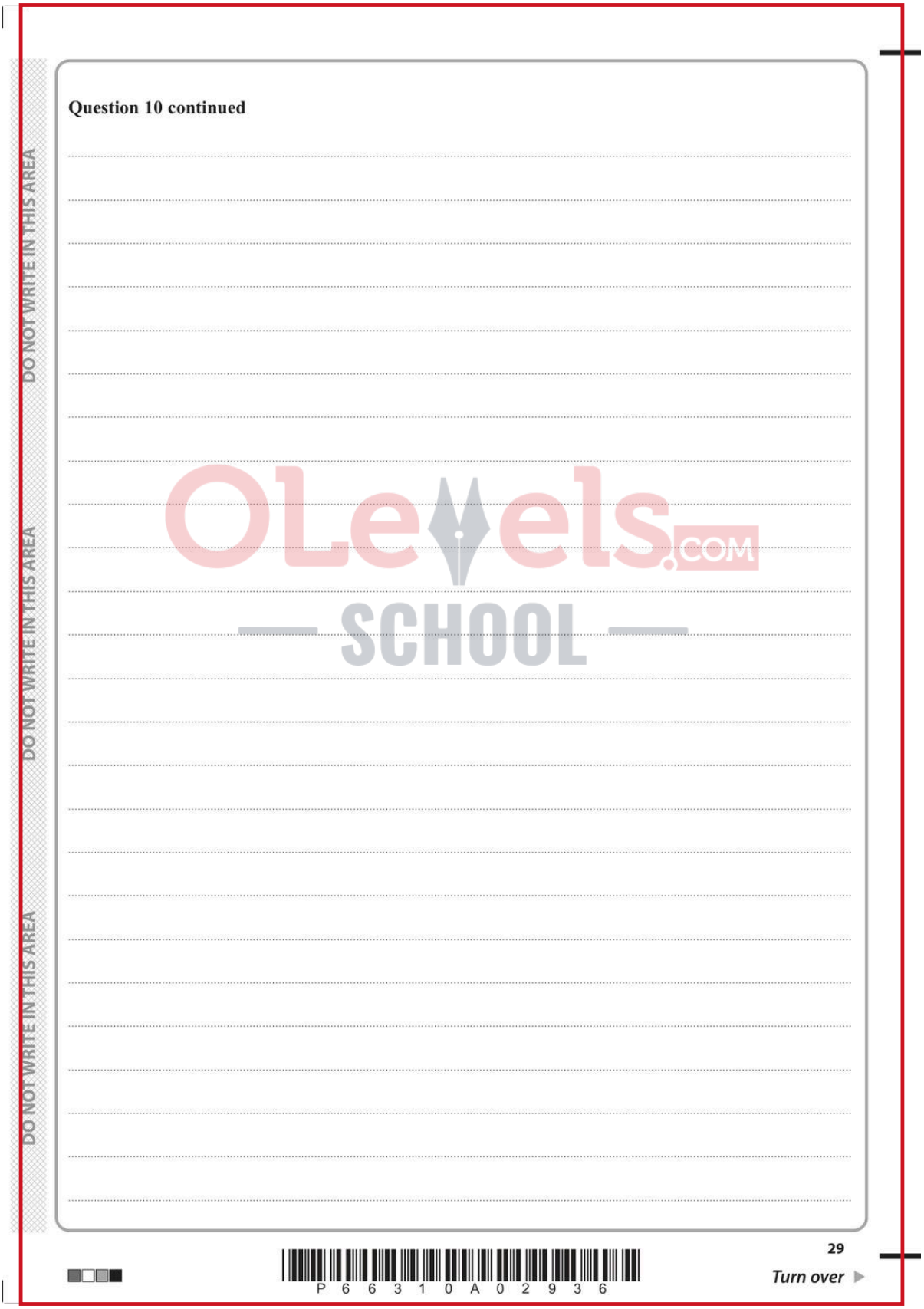
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4. Question 10 · part d · 3 marks

2022 JAN · Paper 01

Further practice · Attempt part d only.



4. Question 10 · part d · 3 marks

2022 JAN · Paper 01

Further practice · Attempt part d only.

Question 10 continued

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4. Question 10 · part d · 3 marks

2022 JAN · Paper 01

Further practice · Attempt part d only.

Question 10 continued

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(Total for Question 10 is 17 marks)

4. Question 10 · part d · 3 marks · Working and answer

Working and answer

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5. Question 6 · part d · 3 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part d only.

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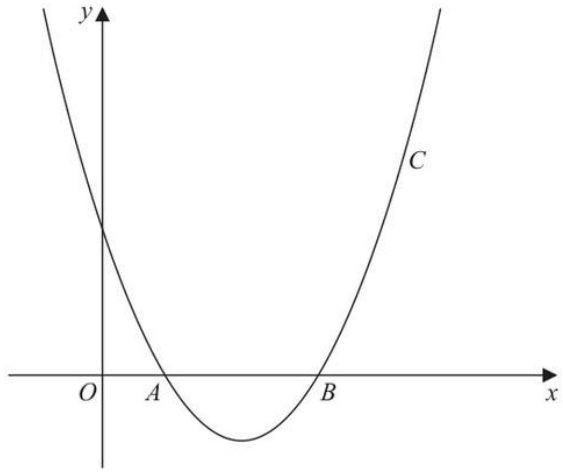


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Figure 3

The curve C with equation $y = x^2 - 5x + 4$ crosses the x-axis at the points A and B, as shown in Figure 3

(a) Find the coordinates of A and the coordinates of B.

(3)

The tangent to C at A meets the tangent to C at B at the point T.

(b) Find the coordinates of T.

(6)

The normal to C at A meets the normal to C at B at the point N.

(c) Find the coordinates of N.

(3)

(d) Find the area of the quadrilateral $ATBN$.

(3)

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5. Question 6 · part d · 3 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 6 continued

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5. Question 6 · part d · 3 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 6 continued

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16

P 6 2 2 8 4 A 0 1 6 3 2

5. Question 6 · part d · 3 marks

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Further practice · Attempt part d only.



Question 6 continued

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(Total for Question 6 is 15 marks)

5. Question 6 · part d · 3 marks · Working and answer

Working and answer

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6. Question 6 · part c · 2 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

6 The curve C has equation

$$y = 4\sqrt{x}$$

The point A on C has coordinates (9, 12)

The tangent to C at the point A meets the x-axis at the point T

(a) Find the coordinates of T

(5)

The normal to C at the point A meets the x-axis at the point N

(b) Find the coordinates of N

(4)

(c) Calculate the area of triangle ATN

(2)

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6. Question 6 · part c · 2 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 6 continued

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(Total for Question 6 is 11 marks)

6. Question 6 · part c · 2 marks · Working and answer

Working and answer

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7. Question 5 · part d · 4 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

5 The line l with gradient $-\frac{1}{12}$ passes through the points A and B with coordinates $(p, 10)$ and $(123, 0)$ respectively.

(a) Show that $p = 3$

(2)

(b) Find an equation for l in the form $rx + sy + t = 0$ where r, s and t are integers.

(2)

The line k is perpendicular to l and passes through the point A.

(c) Find an equation for k in the form $y = mx + c$

(3)

Line k intersects the x-axis at the point C.

(d) Find the exact area of triangle ABC.

(4)

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7. Question 5 · part d · 4 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 5 continued

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7. Question 5 · part d · 4 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 5 continued

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7. Question 5 · part d · 4 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.



Question 5 continued

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(Total for Question 5 is 11 marks)

7. Question 5 · part d · 4 marks · Working and answer

Working and answer

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8. Question 9 · part d · 2 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

9 The point A has coordinates $(-4, 3)$ and the point B has coordinates $(6, 8)$
The points A and B lie on the line k

(a) Find an equation of k

(2)

The point C, on k, is such that $AC : CB = 4:1$

(b) Find the coordinates of point C

(2)

The point D with coordinates (p, q), where $p < 0$, lies on the line l through C that is perpendicular to k

The length of CD is $8\sqrt{5}$

(c) Find the coordinates of D

(6)

(d) Find the area of triangle ACD

(2)

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8. Question 9 · part d · 2 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 9 continued

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8. Question 9 · part d · 2 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 9 continued

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8. Question 9 · part d · 2 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

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Question 9 continued

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(Total for Question 9 is 12 marks)

8. Question 9 · part d · 2 marks · Working and answer

Working and answer

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9. Question 2 · part b · 5 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

2 The point P has coordinates $(-7, 3)$ and the point Q has coordinates $(3, 18)$

The point R divides the line PQ in the ratio $2:3$

The line l passes through R and is perpendicular to the line PQ

(a) Find an equation of line l
Give your answer in the form $ay + bx + c = 0$ where a, b and c are integers.

(5)

The line l crosses the x-axis at the point S

(b) Find the area of triangle QRS

(5)

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9. Question 2 · part b · 5 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 2 continued

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9. Question 2 · part b · 5 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 2 continued

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9. Question 2 · part b · 5 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

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Question 2 continued

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(Total for Question 2 is 10 marks)

9. Question 2 · part b · 5 marks · Working and answer

Working and answer

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10. Question 9 · part c · 3 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

9 The line l_1 with equation $y + 2x - 4 = 0$ passes through the point P with coordinates $(a, 6)$ and the point Q with coordinates $(3, b)$.

(a) Find the value of a and the value of b.

(2)

The line l_2 passes through point P and is perpendicular to l_1

The point R, with coordinates (e, f) lies on l_2 such that $PR = 6\sqrt{5}$

(b) Find the two possible pairs of values of e and f.

(8)

Given that $e < 0$,

(c) find the area of triangle PQR.

(3)

The points P, Q and R lie on a circle C.

(d) Find the coordinates of the centre of C.

(2)

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10. Question 9 · part c · 3 marks · Working and answer

Working and answer

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11. Question 11 · part c · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

11 An equation of the straight line l is $y - 3x = 3$

The point A on l lies on the y-axis.
The point B on l has coordinates $(10, b)$, where b is an integer.
The point C divides AB in the ratio $2:3$

The straight line k passes through C and is perpendicular to l

(a) Show that an equation of k is

$$3y + x - 49 = 0$$

(6)

The point D with coordinates (p, q), where q is positive, is such that AD is parallel to k and the length of AD is $12\sqrt{10}$

(b) Find the coordinates of D

(6)

The point E lies on k such that DE is parallel to the y-axis.
The point F lies on l such that DF is parallel to the y-axis.

(c) Find the exact area of triangle ECF.

(5)

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11. Question 11 · part c · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 11 continued

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11. Question 11 · part c · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 11 continued

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11. Question 11 · part c · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 11 continued

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11. Question 11 · part c · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 11 continued

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(Total for Question 11 is 17 marks)

11. Question 11 · part c · 5 marks · Working and answer

Working and answer

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12. Question 10 · part e · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part e only.

10 The equation of the line L_1 is $y - 3x + a = 0$

The point A with coordinates $(a, 10)$ lies on L_1

(a) Show that $a = 5$

(2)

Line L_1 crosses the y-axis at the point B

(b) Write down the coordinates of B

(1)

The point C with coordinates $(50, -5)$ lies on L_2

Given that L_2 passes through A

(c) (i) show that L_1 and L_2 are perpendicular

(3)

(ii) hence find an equation for L_2 giving your answer in the form $y = px + q$

(2)

The point D has coordinates (m, n)

The length of CD is $15\sqrt{10}$ and the gradient of BD is -3

(d) Find the value of m and the value of n

(6)

(e) Find the area of quadrilateral $ABCD$

(3)

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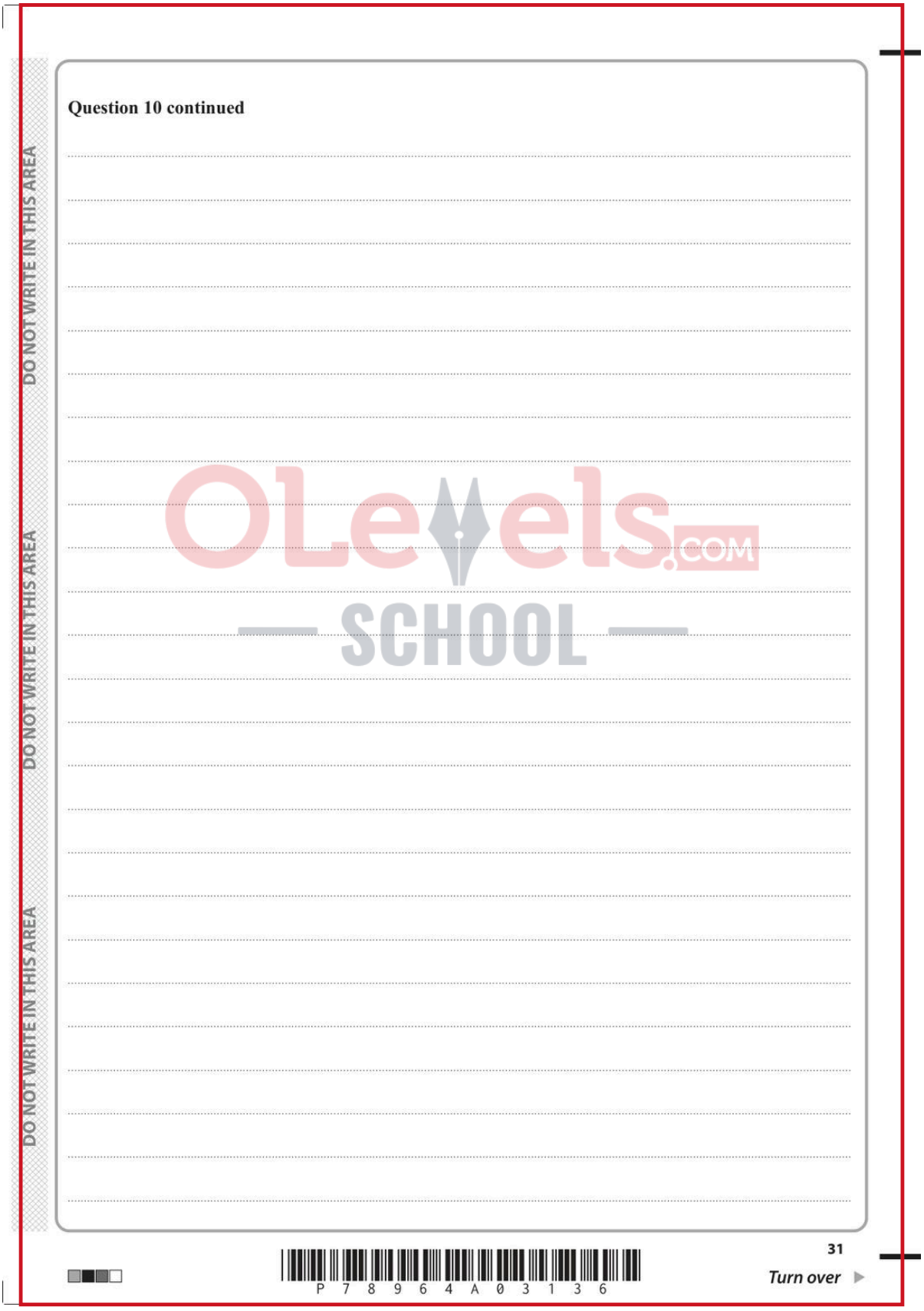
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12. Question 10 · part e · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part e only.



Question 10 continued

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Turn over

12. Question 10 · part e · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part e only.

Question 10 continued

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12. Question 10 · part e · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part e only.

Question 10 continued

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(Total for Question 10 is 17 marks)

12. Question 10 · part e · 3 marks · Working and answer

Working and answer

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13. Question 4 · part e · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part e only.

4 The point A has coordinates $(-4, -10)$ and the point B has coordinates $(3, 11)$
The line l passes through A and B.

(a) Find an equation of l.

(2)

The point P lies on l such that $AP : PB = 3 : 4$

(b) Find the coordinates of P.

(2)

The point Q with coordinates (m, n), where $m < 0$, lies on the line through P that is perpendicular to l.

Given that the length of PQ is $\sqrt{10}$

(c) find the coordinates of Q.

(6)

The point R has coordinates $(-11, -21)$

(d) Show that

(i) AB and RQ are equal in length,

(ii) AB and RQ are parallel.

(4)

(e) Find the area of the quadrilateral $ABQR$.

(2)

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13. Question 4 · part e · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part e only.

Question 4 continued

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13. Question 4 · part e · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part e only.

Question 4 continued

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13. Question 4 · part e · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part e only.



Question 4 continued

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(Total for Question 4 is 16 marks)

13. Question 4 · part e · 2 marks · Working and answer

Working and answer

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14. Question 8 · part e · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.

8 The points A and B have coordinates $(1, 5)$ and $(9, 9)$ respectively.

(a) Find an equation of line AB, giving your answer in the form $ax + by + c = 0$, where a, b and c are integers to be found.

(3)

The line l is perpendicular to AB and passes through the point X which lies on AB such that $AX : XB = 3:1$

(b) Show that an equation of l is $y = -2x + 22$

(5)

The point C has coordinates $(6, p)$

Given that C lies on l

(c) find the value of p

(1)

$ABCD$ is a parallelogram where the x coordinate of D is negative.

(d) Find the coordinates of the point D

(3)

(e) Find the area of the parallelogram $ABCD$

(4)

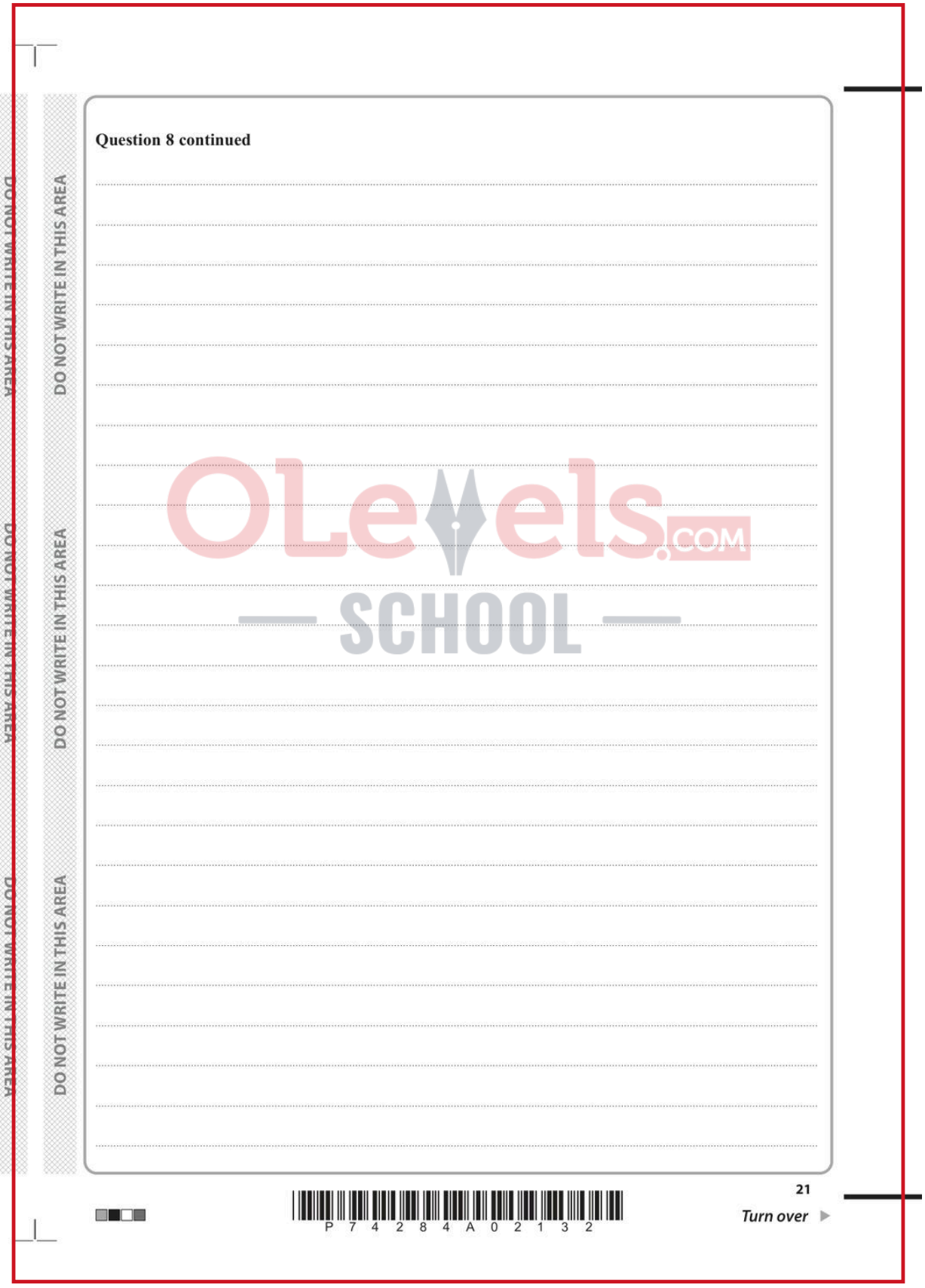
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14. Question 8 · part e · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.



Question 8 continued

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14. Question 8 · part e · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.

Question 8 continued

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14. Question 8 · part e · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.

Question 8 continued

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(Total for Question 8 is 16 marks)

14. Question 8 · part e · 4 marks · Working and answer

Working and answer

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15. Question 11 · part d.ii · 4 marks

2021 NOV · Paper 02

Further practice · Attempt part d.ii only.

11 The points A and B have coordinates $(-3, -5)$ and $(7, 5)$ respectively.

(a) Find an equation for the line AB

(2)

The point C has coordinates $(p, 1)$ where $p < 0$

Given that AC and BC are perpendicular,

(b) prove that $p = -5$

(7)

The point D, where BCD is a straight line, is such that C divides BD in the ratio $4:3$

(c) Find the coordinates of D

(2)

(d) (i) Find the exact length of AC

(ii) Hence, or otherwise, find the area of triangle ABD

(4)

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15. Question 11 · part d.ii · 4 marks

2021 NOV · Paper 02

Further practice · Attempt part d.ii only.

Question 11 continued

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15. Question 11 · part d.ii · 4 marks

2021 NOV · Paper 02

Further practice · Attempt part d.ii only.

Question 11 continued

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(Total for Question 11 is 15 marks)

15. Question 11 · part d.ii · 4 marks · Working and answer

Working and answer

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16. Question 8 · part e · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part e only.

8 The points A and B have coordinates $(-6, 8)$ and $(12, 2)$ respectively.

(a) Find an equation of the straight line passing through A and B in the form $ax + by + c = 0$, where a, b and c are integers to be found.

(3)

(b) Find the exact length of AB

(2)

The point X with coordinates (m, n) lies on AB such that $AX:XB = 1:2$

(c) Find the value of m and the value of n

(2)

The line L passes through the point X and is perpendicular to AB
The point C with coordinates (p, q) lies on L where $p > 0$ and $q > 0$

Given that AB is a diameter of a circle and C also lies on the circumference of the circle,

(d) find

(i) the exact value of p

(ii) the exact value of q

(7)

(e) Find the exact area of triangle ABC

(3)

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16. Question 8 · part e · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part e only.

Question 8 continued

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16. Question 8 · part e · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part e only.

Question 8 continued

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16. Question 8 · part e · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part e only.

Question 8 continued

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(Total for Question 8 is 17 marks)

16. Question 8 · part e · 3 marks · Working and answer

Working and answer

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17. Question 3 · part b · 5 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

3 The point A has coordinates $(1, 7)$ and the point B has coordinates $(9, 3)$

The line l is the perpendicular bisector of AB

(a) Find an equation of l

(5)

The line l crosses the x-axis at the point C

(b) Find the area of the triangle ABC

(5)

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17. Question 3 · part b · 5 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 3 continued

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(Total for Question 3 is 10 marks)

17. Question 3 · part b · 5 marks · Working and answer

Working and answer

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